

Numerical Integration of the Nonlinear Pendulum

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The nonlinear pendulum is a classical mechanical system whose equation of motion, $\ddot{\theta} + (g/L)\sin\theta = 0$, is a second-order nonlinear ordinary differential equation. Its exact solution can be written in terms of elliptic integrals and functions, which give the complete time evolution for any initial condition. Numerical integration is nonetheless essential, because it extends directly to more realistic dynamics, such as damping or driving, for which no closed-form solution exists. The exact solution then serves as a rigorous benchmark for the numerical methods. In this letter, I integrate the equation of motion with the explicit Euler, symplectic Euler, and fourth-order Runge-Kutta (RK4) methods. I verify each method against two physical properties: conservation of the total energy, and agreement with the exact period $T = 4K(k^2)/\omega_0$, where $k^2 = E/(2mgL)$ and K is the complete elliptic integral. Explicit Euler expands phase-space area, so its energy grows steadily until the numerical pendulum crosses the separatrix and begins to rotate. Symplectic Euler preserves phase-space area, which keeps its energy error bounded, while RK4 exhibits only a negligible slow drift. The measured period errors converge at orders 1, 2, and 4, respectively; symplectic Euler attains second-order accuracy in the period because its angle sequence is identical to that of the Störmer-Verlet method. The numerical periods reproduce both the small-angle limit $T \rightarrow 2\pi\sqrt{L/g}$ and the growth of the period with amplitude. I then use energy conservation to reduce the motion to a definite integral for the time required to swing from the bottom to a given angle, and a change of variables removes the integrable singularity at the turning point. I evaluate this integral using left Riemann sums, the trapezoidal rule, and Simpson's rule, which converge at orders 1, 2, and 4 and agree exactly with the corresponding SciPy implementations. The integral recovers the small-angle limit and agrees with independent RK4 trajectories to within 10^{-8} s. Four times the quarter-period integral reproduces the exact period from the ODE to ten significant digits, demonstrating that the equation of motion and energy conservation describe the same dynamics.

I. INTRODUCTION

A pendulum of mass m on a rigid rod of length L obeys

$$\ddot{\theta} + \omega_0^2 \sin\theta = 0, \quad \omega_0 = \sqrt{g/L}. \quad (1)$$

For small angles, $\sin\theta \approx \theta$ and the motion is simple harmonic with period $T_0 = 2\pi/\omega_0$, independent of amplitude. At larger amplitudes the restoring torque is weaker than linear, so the period grows, diverging as $\theta_0 \rightarrow \pi$, where the pendulum approaches its unstable inverted equilibrium. Because the system is conservative, the energy

$$E = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos\theta) \quad (2)$$

is constant, and for a pendulum swinging back and forth ($E < 2mgL$) the exact period is

$$T = \frac{4}{\omega_0}K(k^2), \quad k^2 = \frac{E}{2mgL} = \sin^2 \frac{\theta_0}{2}, \quad (3)$$

with K the complete elliptic integral of the first kind. The energy $E = 2mgL$ defines the separatrix between swinging and full rotation. Equations (2) and (3) provide exact benchmarks for the numerical methods below.

II. METHODS

A. ODE integration

Writing $\omega = \dot{\theta}$ and $a(\theta) = -\omega_0^2 \sin\theta$, Eq. (1) becomes a first-order system, which I advance with step h using

$$\begin{aligned} \text{Explicit Euler: } \theta_{n+1} &= \theta_n + h\omega_n, & \omega_{n+1} &= \omega_n + h a(\theta_n), \\ \text{Symplectic Euler: } \omega_{n+1} &= \omega_n + h a(\theta_n), & \theta_{n+1} &= \theta_n + h\omega_{n+1}, \end{aligned}$$

and classical RK4. All runs use $m = g = L = 1$ and $\theta_0 = \text{rad}$ released from rest, giving $E_0 =$ (in units of mgL) and $T =$, compared with $T_0 =$ (in units of $1/\omega_0$). The numerical period is the mean time between upward zero crossings of θ , each located on a cubic Hermite interpolant built from θ and ω ; linear interpolation would otherwise limit the measured accuracy to second order.

B. Definite integral

Solving Eq. (2) for $\dot{\theta}$ gives the time to swing from $\theta = 0$ to θ_1 . The substitution $\sin(\theta/2) = k \sin\phi$ removes the singularity at the turning point:

$$t(\theta_1) = \frac{1}{\omega_0} \int_0^{\phi_1} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \frac{F(\phi_1, k^2)}{\omega_0}, \quad \sin\phi_1 = \frac{\sin(\theta_1/2)}{k} \quad (4)$$

where F is the incomplete elliptic integral of the first kind; $\phi_1 = \pi/2$ gives the quarter period $T/4$. I evaluate

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Eq. (4) on N equal intervals with the left Riemann sum, the trapezoidal rule, and Simpson's rule.

III. VERIFICATION

A. ODE: energy conservation

Figure 1 shows the energy for each method. Explicit Euler expands phase-space area, so its energy grows until it crosses the separatrix ($E = 2$) at $t \approx 17.7$ and the pendulum begins to rotate. Symplectic Euler conserves a nearby modified energy, so its error oscillates within ± 0.03 without drifting. RK4's drift is invisible on this scale. The maximum energy error scales as h for both Euler methods and roughly as h^4 – h^5 for RK4.

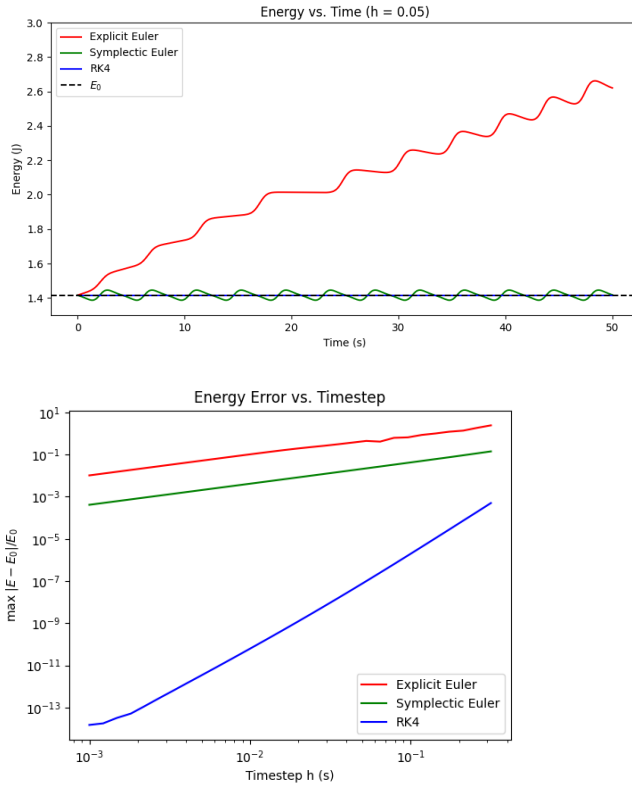


FIG. 1. Top: energy vs. time ($h = 0.05$). Bottom: maximum relative energy error vs. step size.

B. ODE: period

Figure 2 compares the measured period with Eq. (3). All methods recover $T \rightarrow T_0$ at small amplitude and the growth of T with amplitude. Explicit Euler lies above the exact curve, because its growing energy widens the swing, and its last point is missing because that run crossed the separatrix. The period errors converge at orders 1, 2, and 4. Symplectic Euler is second order here because its

θ sequence satisfies $\theta_{n+1} - 2\theta_n + \theta_{n-1} = h^2 a(\theta_n)$, the Störmer–Verlet recursion. RK4 reaches machine precision at small h , and its dip near $h \approx 0.1$ is a sign change in the error.

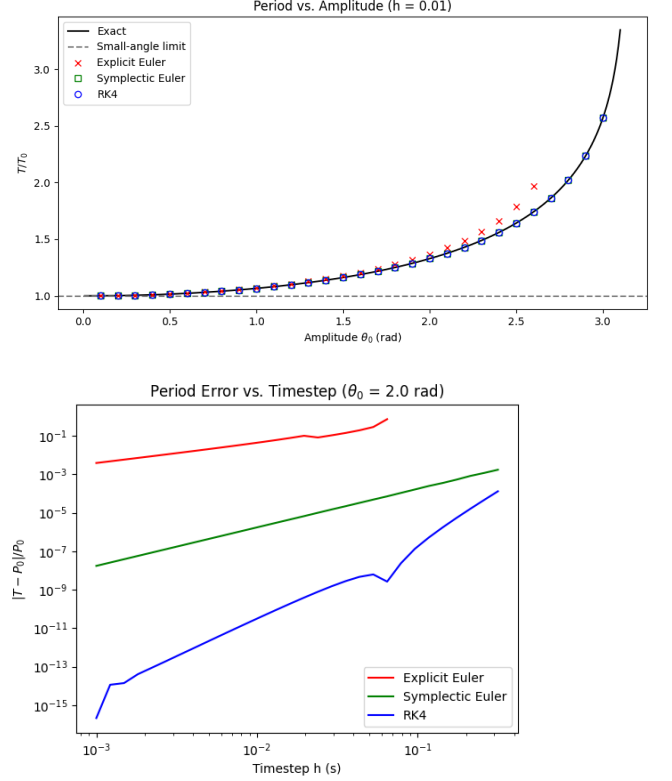


FIG. 2. Top: period vs. amplitude ($h = 0.01$) against the exact result and the small-angle limit. Bottom: relative period error vs. step size at $\theta_0 = \text{rad}$.

C. Integral: small-angle limit

As $\theta_0 \rightarrow 0$, $k \rightarrow 0$ and the integrand in Eq. (4) approaches a constant, so $4t(\pi/2) \rightarrow T_0$. Figure 3 confirms this limit, the leading correction $T/T_0 \approx 1 + \theta_0^2/16$, and convergence orders of 1, 2, and 4, with SciPy agreeing exactly. The left Riemann sum falls below the exact curve at large amplitude because the integrand increases toward the turning point. Over the full quarter period the trapezoidal and Simpson results are accurate even at $N = 16$, because the integrand is symmetric about both endpoints and the leading error terms cancel.

D. Integral: consistency with the ODE

Launching RK4 from $\theta = 0$ with energy E_0 places it on the same phase-space orbit. Its crossing times agree with Eq. (4) to within 1.4×10^{-8} (Fig. 4), consistent with RK4's $O(h^4)$ error at $h = 0.01$, and $4t(\pi/2)$ reproduces

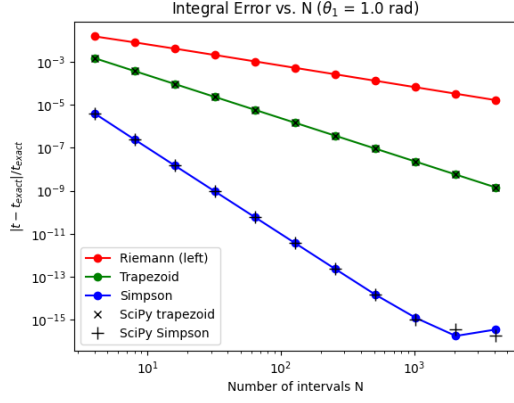


FIG. 3. Top: relative error of $t(\theta_1)$ vs. number of intervals. Bottom: $4t(\pi/2)/T_0$ vs. amplitude ($N = 16$).

$T =$ to ten significant digits. The curve steepens near θ_0 as the pendulum slows to its turning point.

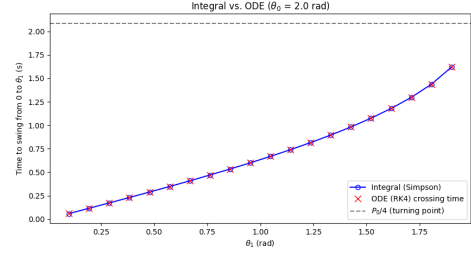


FIG. 4. Time to swing from 0 to θ_1 : Simpson integral vs. RK4 crossing times.

IV. CONCLUSION

I solved the nonlinear pendulum both as an ODE and as a definite integral, and validated each method against the exact elliptic-integral solution. For the ODE, the energy and period tests distinguish the methods by their geometric structure. Explicit Euler violates conservation of phase-space area and performs noticeably worse than the exact solution, eventually crossing the separatrix. Symplectic Euler keeps the energy error bounded and achieves second-order accuracy in the period. RK4 is the most accurate method at every step size tested. For the integral, the Riemann, trapezoidal, and Simpson rules converge at orders 1, 2, and 4 and match SciPy exactly, and the integral reproduces the period from the ODE to ten significant digits.

The observed shortcomings suggest clear improvements. For long simulations, explicit Euler should be replaced by a symplectic scheme, and a higher-order symplectic integrator would remove RK4's slow energy drift. The left Riemann sum's underestimate near the turning point could be reduced with the midpoint rule. For amplitudes near π , where the integrand becomes sharply peaked, adaptive quadrature would be more efficient than a uniform grid.

Due to limits on time and effort, several parts of the assignment were not completed. I did not compare the ODE solutions against a SciPy integrator such as `solve_ivp`, and I did not plot example trajectories $\theta(t)$ against the exact solution in terms of Jacobi elliptic functions. I also did not derive the local and global truncation errors of each method. The convergence plots show the expected global error scaling, but I did not separately measure the local (per-step) error.